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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/965,442	01/22/2026	Eric Haywood	

CONFIRMATION NO. 3409

FORMALITIES LETTER

OC000000102063868

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

Date Mailed: 02/26/2026

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to the above-identified provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee has not been received.
- Surcharge as set forth in 37 CFR 1.16(g) has not been received.

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SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- $**325** basic filing fee.
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/nhnguyen/

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY.DOCKET.NO	TOT CLAIMS	IND CLAIMS
63/965,442	01/22/2026		0			

CONFIRMATION NO. 3409

FILING RECEIPT

OC000000102063865

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

Date Mailed: 02/26/2026

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

Eric Haywood, Fort Collins, CO;

Applicant(s)

HeadySystems Inc., Fort Collins, CO;

Power of Attorney: None

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/25/2026

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 63/965,442**

Projected Publication Date: None

Non-Publication Request: No

Early Publication Request: No

Title

Microgravity Resonance Stabilization via Spectral Mode Identification and Phase-Locked Counteractuation (HeadySpace)

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 221639

Thu Feb 26 05:12:05 EST 2026

Dear Patent Center Customer:

Correspondence Address:

Eric Haywood
149 Remington Street Apt 425
Fort Collins,COLORADO,80524
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 221639, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

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Application	Document	Mailroom Date	Attorney Docket No.
63965442	NTC.MISS.PRT	02/26/2026	
63965442	APP.FILE.REC	02/26/2026	
63965442	WELCOME.LET	02/26/2026	

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Sincerely,

The Patent Center Team

Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

Residence Information
US Residency
Fort Collins, CO
UNITED STATES

Mailing Address
149 Remington Street Apt 425
Fort Collins, CO 80524
UNITED STATES

Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**Microgravity Resonance Stabilization via Spectral Mode Identification and Phase-Locked Counteractuation (HeadySpace)**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

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Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
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A. **Priority Document Exchange (PDX)**
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The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

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☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:
Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 1

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

1. **HeadySystems Inc.**
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

Phone: 9704445112
Fax: ---
Email: admin@headysystems.com

Applicant	Organization
Applicant type	Assignee

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

9704445112
---
admin@headysystems.com

Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	01/22/2026

PROVISIONAL PATENT APPLICATION

Microgravity Resonance Stabilization via Spectral Mode Identification and Phase-Locked Counteractuation (HeadySpace)

(Specification and Drawings)

Applicant / Assignee	HeadySystems Inc.
Inventor	Eric Haywood
Correspondence Address	149 Remington Street Unit 425, Fort Collins, CO 80524
Entity Status	Small Entity
Government Interest	None
Attorney Docket	PPA-030
Date	January 22, 2026

Title

[0001] PPA-030: HeadySpace — Microgravity Resonance Stabilization via Spectral Mode Identification and Phase-Locked Counteractuation

Abstract

[0002] A microgravity stabilization and manufacturing quality system treats structural vibration, actuator currents, and control telemetry as waveform data. A spectral estimator identifies dominant vibration modes and cross-coupled interference among onboard subsystems (e.g., pumps, reaction wheels, compressors, tool heads, and computing workloads). A resonant null controller computes phase-locked counteractuation signals that drive active dampers or other actuators to cancel or reduce excitation of structural modes, preserving microgravity quality and improving process precision. A harmonic scheduling layer further phase-aligns periodic software and electromechanical operations to avoid exciting sensitive frequencies. The system provides a microgravity quality metric and a safety envelope that bounds actuation to maintain stability and protect hardware.

Technical Field

[0003] The disclosure relates to spacecraft and microgravity systems, including vibration suppression, spectral analysis, active damping, control theory, cyber-physical scheduling, and quality assurance for microgravity manufacturing and scientific payload operations.

Background

[0004] Microgravity manufacturing and scientific payloads are sensitive to low-amplitude vibrations generated by onboard systems such as pumps, fans, compressors, reaction wheels, robotic actuators, and periodic compute workloads.

[0005] In microgravity environments, small oscillatory forces can accumulate to degrade process precision, introduce defects, reduce measurement fidelity, and impair docking or pointing stability. Conventional vibration mitigation approaches are often static (passive damping) or tuned for a limited set of operating conditions and may not adapt to changing configurations, payload mass, or aging hardware.

[0006] There is a need for a runtime control system that detects evolving vibration modes in the frequency domain and applies phase-locked counteractuation and scheduling adjustments to maintain a stable microgravity environment while respecting safety constraints and mission operations.

Summary

[0007] A multi-sensor telemetry layer collects vibration and structural signals (e.g., accelerometers, IMUs, strain gauges) together with actuator and compute telemetry and timestamps the streams using a stable timebase.

[0008] A spectral estimator computes frequency-domain representations and identifies structural resonance modes, cross-coupled interference, and coherence among subsystems.

[0009] A resonant null controller selects one or more modes to suppress and drives active actuators using phase-locked counteractuation signals that reduce vibration energy in targeted bands while maintaining stability.

[0010] A harmonic scheduling layer optionally micro-delays periodic operations (software tasks and electromechanical cycles) to avoid exciting sensitive frequencies and to maintain constructive alignment with damping capacity.

[0011] A microgravity quality metric and dissonance alarm provide monitoring, fault prediction, and evidence for manufacturing quality and maintenance.

Brief Description of the Drawings

[0012] - FIG. 1 illustrates an example microgravity resonance control architecture including sensors, spectral estimation, mode management, scheduling, and active actuation.

[0013] - FIG. 2 illustrates an example reduction in vibration energy at dominant structural resonance modes.

Detailed Description

System Overview

[0014] In one embodiment, the system is deployed on a spacecraft, station module, free-flyer manufacturing platform, or microgravity laboratory. The system may run on an onboard computer and communicate with actuators over deterministic fieldbuses.

[0015] The system supports multiple operating modes including a learning mode (mode identification), a control mode (active cancellation), and a safety mode (bounded actuation and rollback).

Telemetry Acquisition and Timebase

[0016] Telemetry sources include inertial sensors, structural strain sensors, and motor electrical signals. Control telemetry includes commanded actuator setpoints and observed positions or currents. Software telemetry includes periodic task schedules and CPU wakeups that can excite structural modes through power and mechanical coupling.

[0017] In one embodiment, telemetry is synchronized to a monotonic timebase and resampled to a uniform cadence. Time synchronization may use an authenticated onboard clock source to preserve phase coherence across sensors.

Spectral Mode Identification

[0018] The spectral estimator computes power spectral density and cross-spectral density across sensor channels to estimate dominant frequencies and mode shapes. Short-time Fourier transforms or wavelet transforms may be used to capture changing modes as payload configuration changes.

[0019] Mode identification may include tracking of resonance frequency drift, damping ratios, and coupling coefficients between actuators and measured structural responses.

Resonant Null Control and Phase-Locked Counteractuation

[0020] For a targeted resonance mode, the controller computes a counteractuation signal that is phase-shifted relative to the measured vibration such that injected energy is

minimized and damping is maximized. The controller may implement a phase-locked loop to maintain phase alignment as the mode frequency drifts.

[0021] Actuators may include active vibration isolators, voice-coil dampers, reaction wheels, control-moment gyros, pump speed controllers, or tool-head actuators. The controller can distribute counteractuation across multiple actuators to reduce local loads and to satisfy actuator saturation constraints.

Harmonic Scheduling Layer

[0022] In addition to physical counteractuation, the system may reduce excitation by micro-delaying periodic operations. Examples include staggering pump cycles, aligning compressor starts, shifting high-current compute bursts, and coordinating tool-path segments in manufacturing.

[0023] Scheduling adjustments are bounded to preserve deadlines and mission constraints. The system may maintain a constraint set specifying maximum allowable jitter per task and maximum rate-of-change of offsets.

Microgravity Quality Metric and Evidence

[0024] The system computes a microgravity quality metric Q derived from integrated vibration energy over sensitive bands, coherence among channels, and persistence of modes. Q may be recorded alongside manufacturing batch identifiers to provide evidence of environmental quality during a process.

[0025] A dissonance alarm triggers when the system detects anomalous spectral signatures consistent with imbalance, bearing wear, loose mounts, or controller instability.

Safety Envelope and Fault Handling

[0026] The system enforces a safety envelope that bounds actuation magnitude, frequency content, and slew rate. If instability is detected, the system reverts to a safe mode that disables active cancellation and returns to passive damping configurations.

[0027] In one embodiment, all control decisions and applied offsets are logged with timestamps for audit and post-mission analysis.

Example Claims (Non-Limiting)

[0028] 1. A computer-implemented method comprising: collecting vibration telemetry from a plurality of sensors on a microgravity platform; performing spectral analysis to identify at least one structural resonance mode; computing a phase-locked counteractuation signal for the resonance mode; and driving at least one actuator with the

counteractuation signal to reduce vibration energy in a frequency band associated with the resonance mode.

[0029] 2. The method of claim 1, further comprising maintaining phase alignment using a phase-locked loop responsive to drift in the resonance mode frequency.

[0030] 3. The method of claim 1, wherein the actuator comprises at least one of an active vibration damper, reaction wheel, pump controller, compressor controller, or tool-head actuator.

[0031] 4. The method of claim 1, further comprising applying a bounded micro-delay to a periodic operation to reduce excitation of the resonance mode while preserving task correctness.

[0032] 5. The method of claim 1, further comprising computing a microgravity quality metric derived from integrated vibration energy and storing the metric as evidence for a manufacturing or scientific operation.

[0033] 6. The method of claim 1, further comprising enforcing a safety envelope that bounds actuation magnitude and disables active control upon detection of instability.

Drawings

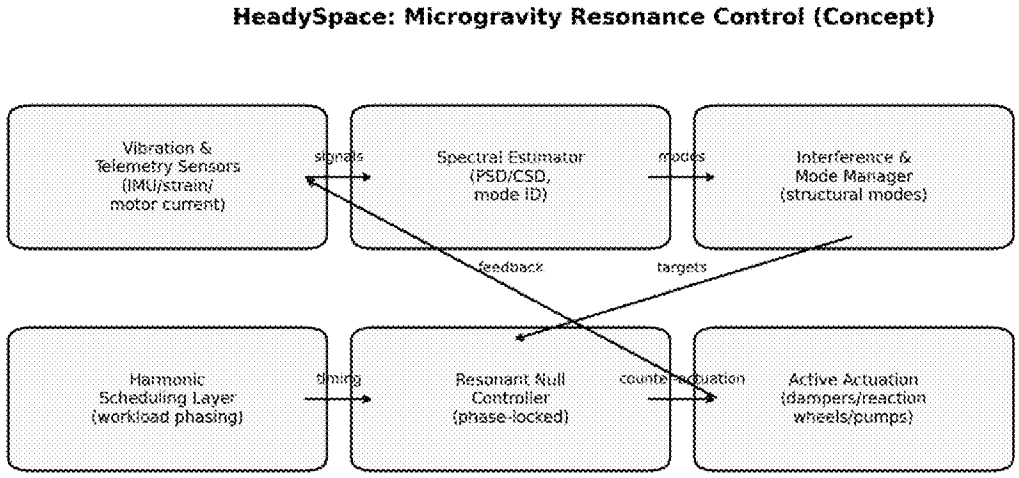

FIG. 1 — Microgravity resonance control architecture.

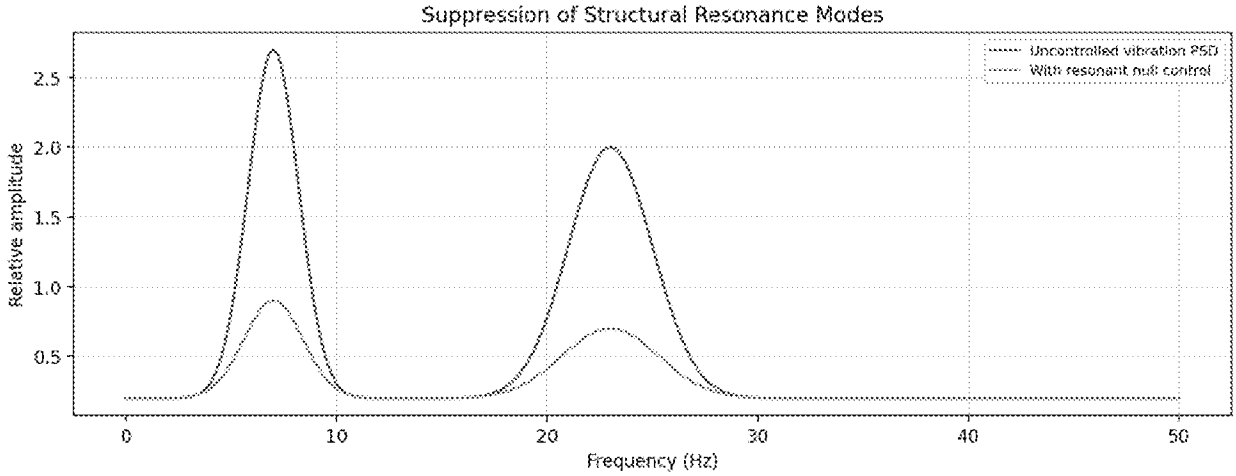

FIG. 2 — Reduction of vibration energy at dominant resonance modes.

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/965,442	**01/22/2026 05:34:59 AM Z ET**	**-**

Title of Invention

Microgravity Resonance Stabilization via Spectral Mode Identification and Phase-Locked Counteractuation (HeadySpace)

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	3409	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74085734	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74085734.pdf	5	Application Data Sheet	111 KB
PPA-030_HeadySpace_SMALL_ENTITY_FINAL-APP.TEXT.docx	6	Application body structured text document	265 KB

Warning: The specification paragraph numbering is not consecutive. Please review and revise if necessary.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)

generatedADS74085734.pdf	E3963AB66AE4338E573AE396A5B4829801CE3F5CC8A1BEEFD727B6F4ABFA8D3C154A3205B5785BB61400028C32E8BC3DE66F83505EF3F4F5955B425D883A4734
PPA-030_HeadySpace_SMALL_ENTITY_FINAL-APP.TEXT.docx	6F5AAA121AAEDDA124DCE6A4654B4B3D6DF29AD5CD202E60BD24F70A373610A31667F41D4254D68F31E6BAAE88E0D3ADF3873B258E36E93AC99953D029FF283B

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Application Number: 63965442 Document Date: 01/22/2026

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